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OFFICIAL ACADEMIC STUDY HANDBOOK & COURSE MANUAL

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PYTHON FOR DATA SCIENCE

Python for Data Science: Analytics & Machine Learning

High-Performance Numerical Computing with NumPy, Pandas, Scikit-Learn & Matplotlib

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CHAPTER 1

Vectorized Computing with NumPy & Array Manipulation

1. Theoretical Foundations & Scope

NumPy forms the fundamental building block for scientific computing in Python. This chapter explores ndarrays, memory strides, vectorized operations, broadcasting semantics, and linear algebra routines.

2. Core Conceptual Framework

- **Array Broadcasting:** Mechanisms governing element-wise operations on arrays of differing compatible shapes without explicit loops.
- **Memory Layout & Slicing:** C-contiguous versus Fortran-contiguous orders, zero-copy memory views, and strided indexing.
- **Matrix Decomposition:** Eigenvalue computation, singular value decomposition (SVD), and matrix inverses in `numpy.linalg`.

Key Practical Insights & Examination Takeaways:

- ' Eliminate slow Python loops using vectorized universal functions (ufuncs).
- ' Master boolean masking and fancy indexing for multi-dimensional filtering.
- ' Optimize memory footprints using appropriate int/float dtypes.

CHAPTER 2

Data Wrangling, Aggregation & Pipelines with Pandas

1. Theoretical Foundations & Scope

Pandas provides intuitive, high-performance data structures for structured data manipulation. We cover Series, DataFrames, multi-indexing, handling missing values, groupby aggregations, and tidy data reshaping.

2. Core Conceptual Framework

- **DataFrame Indexing:** Precision differences between loc (label-based) and iloc (integer position-based) selection.
- **Groupby Mechanics:** The Split-Apply-Combine paradigm for parallel statistical computations across categorical segments.
- **Missing Data Imputation:** Techniques for identifying NaN values and applying forward-fill, mean/median imputation, or drop criteria.

Key Practical Insights & Examination Takeaways:

- ' Perform complex database-style merges, joins, and concatenations.
- ' Reshape tabular datasets using pivot_table and melt operations.
- ' Parse and resample time-series data using date range indices.

CHAPTER 3

Predictive Modeling & Evaluation with Scikit-Learn

1. Theoretical Foundations & Scope

Scikit-Learn provides a clean, unified API for supervised and unsupervised machine learning. We examine linear models, decision trees, random forests, feature scaling, cross-validation, and performance metrics.

2. Core Conceptual Framework

- **Transformer-Estimator API:** Consistent fit, transform, and predict method signatures enabling robust pipeline construction.
- **Cross-Validation & Grid Search:** K-fold stratified cross-validation and hyperparameter optimization preventing data leakage.
- **Evaluation Metrics:** ROC-AUC curves, precision-recall trade-offs, confusion matrices, and F1-score interpretation.

Key Practical Insights & Examination Takeaways:

- ' Build end-to-end preprocessing pipelines using ColumnTransformer.
- ' Diagnose bias versus variance trade-offs using learning curves.
- ' Select optimal classification and regression metrics for imbalanced datasets.

Academic References & Further Study

[1] Quality Education Academic Standards & Pedagogical Guidelines, 2026 Edition.

[2] International Computer Science & Engineering Curriculum Framework.

[3] Educational Technology & Assessment Methodology Research Series.

[4] Open Access Educational Resource Repository - Quality Education Platform.

